

MIHIR KULKARNI

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EDUCATION

- Ph.D. in Engineering Cybernetics** *January 2022 - August 2025*
Norwegian University of Science and Technology
Department of Engineering Cybernetics
Dissertation: [Vision-based Navigation for Aerial Robots: From Parallelized Simulation to Resilient Flight in Cluttered Environments.](#)
Advisors: [Prof. Dr. Kostas Alexis](#) (*supervisor*), [Prof. Dr. Davide Scaramuzza](#) (*co-supervisor*)
Committee: Prof. Dr. Georgia Chalvatzaki, Prof. Dr. Dimitrios Kanoulas, Prof. Dr. Damiano Varagnolo
- M.S. in Computer Science and Engineering** *August 2020 - December 2021*
University of Nevada, Reno
Department of Computer Science and Engineering
- B.E. in Mechanical Engineering** *August 2016 - July 2020*
Birla Institute of Technology and Science, Pilani (Goa Campus)
Department of Mechanical Engineering

EXPERIENCE

Researcher *Sep 2025 - Present*
Department of Engineering Cybernetics, NTNU *Trondheim, Norway*

- Co-led the development of the [Unified Autonomy Stack](#), enabling autonomy across diverse platforms.
- Built support for multirotor platforms for [NVIDIA Isaac Lab](#).
- Designed and fabricated [UniPilot](#), a hardware-software autonomy module for diverse platforms.
- Building foundation control/navigation policies transferable across arbitrary aerial platforms.

PhD Candidate *Jan 2022 - Aug 2025*
Department of Engineering Cybernetics, NTNU *Trondheim, Norway*

- Created and open-sourced the [Aerial Gym Simulator](#) (*Stars: 675+, Forks: 100*). Control and navigation policies bridging the sim2real gap can be trained in minutes.
- Developed and field tested [vision-driven RL policies](#) for navigation in densely cluttered environments.
- Developed and field-tested uncertainty-aware methods for GNSS-denied autonomous navigation.

Graduate Research Assistant *Jan 2021 - Dec 2021*
Department of Computer Science and Engineering, UNR *Reno, USA*

- Developed [COHORT](#) a heterogeneous multi-robot exploration path planning framework.
- Led the artifact detection sub-system for Team CERBERUS during the [DARPA Subterranean\(SubT\) Challenge](#), correctly detecting the highest number of artifacts.
- Performed system integration, hardware customization and software integration for aerial and wheeled systems for the SubT Challenge.

AWARDS AND ACHIEVEMENTS

- Winner – [DARPA Subterranean Challenge](#)** Final Event (USD 2mn prize) *2021*
- Outstanding Deployed Systems Paper Award – IEEE ICRA** *2022*
- Best Paper Award – IEEE ICAR** *2023*
- Certificate of Special Congressional Recognition – United States Senate** *2021*

JOURNAL PUBLICATIONS

1. M. Dharmadhikari, N. Khedekar, **M. Kulkarni**, M. Nissov, M. Jacquet, A. Zacharia, M. Harms, A. G. Puigjaner, P. Weiss, K. Alexis, **“The Unified Autonomy Stack: Toward a Blueprint for Generalizable Robot Autonomy”**. *Under Review*. [DOI](#).
2. W. Rehberg, **M. Kulkarni**, P. Weiss and K. Alexis, **“Efficient Knowledge Transfer for Jump-Starting Control Policy Learning of Multirotors through Physics-Aware Neural Architectures”**, IEEE Robotics and Automation Letters. [DOI](#).
3. G. Malczyk, **M. Kulkarni** and K. Alexis, **“Semantically-Driven Deep Reinforcement Learning for Inspection Path Planning,”**, IEEE Robotics and Automation Letters. [DOI](#).
4. **M. Kulkarni**, W. Rehberg and K. Alexis, **“Aerial Gym Simulator: A Framework for Highly Parallelized Simulation of Aerial Robots”**, IEEE Robotics and Automation Letters. [DOI](#).
5. M. Tranzatto, M. Dharmadhikari, L. Bernreiter, M. Camurri, S. Khattak, F. Mascarich, P. Pfrendschuh, D. Wisth, S. Zimmermann, **M. Kulkarni**, V. Reijgwart, B. Casseau, T. Homberger, P. De Petris, L. Ott, W. Tubby, G. Waibel, H. Nguyen, C. Cadena, R. Buchanan, L. Wellhausen, N. Khedekar, O. Andersson, L. Zhang, T. Miki, T. Dang, M. Mattamala, M. Montenegro, K. Meyer, X. Wu, A. Briod, M. Mueller, M. Fallon, R. Siegwart, M. Hutter, K. Alexis, **“Team CERBERUS Wins the DARPA Subterranean Challenge: Technical Overview and Lessons Learned”**, Field Robotics. [DOI](#).
6. M. Tranzatto, T. Miki, M. Dharmadhikari, L. Bernreiter, **M. Kulkarni**, F. Mascarich, O. Andersson, S. Khattak, M. Hutter, R. Siegwart, K. Alexis, **“CERBERUS in the DARPA Subterranean Challenge”** Science Robotics. [DOI](#).
7. F. Mascarich, **M. Kulkarni**, P. de Petris, T. Wilson, K. Alexis, **“Autonomous Mapping and Spectroscopic Analysis of Distributed Radiation Fields using Aerial Robots”**, Autonomous Robots. [DOI](#).
8. M. Tranzatto, F. Mascarich, L. Bernreiter, C. Godinho, M. Camurri, S. Khattak, T. Dang, V. Reijgwart, J. Loje, D. Wisth, S. Zimmermann, H. Nguyen, M. Fehr, L. Solanka, R. Buchanan, M. Bjelonic, N. Khedekar, M. Valceschini, F. Jenelten, M. Dharmadhikari, T. Homberger, P. De Petris, L. Wellhausen, **M. Kulkarni**, T. Miki, S. Hirsch, M. Montenegro, C. Papachristos, F. Tresoldi, J. Carius, G. Valsecchi, J. Lee, K. Meyer, X. Wu, J. Nieto, A. Smith, M. Hutter, R. Siegwart, M. Mueller, M. Fallon, K. Alexis, **“CERBERUS: Autonomous Legged and Aerial Robotic Exploration in the Tunnel and Urban Circuits of the DARPA Subterranean Challenge”**, Field Robotics. [DOI](#).

CONFERENCE PUBLICATIONS

1. G. Malczyk, **M. Kulkarni**, K. Alexis **“Reinforcement Learning for Active Perception in Autonomous Navigation”**. IEEE International Conference on Robotics and Automation (ICRA) 2026. [DOI](#).
2. E. Arza, W. Rehberg, P. Weiss, **M. Kulkarni**, K. Alexis **“Performance-guided Task-specific Optimization for Multirotor Design.”** *Under Review*. [DOI](#).
3. **M. Kulkarni**, M. Dharmadhikari, N. Khedekar, M. Nissov, M. Singh, P. Weiss and K. Alexis, 2025. **“UniPilot: Enabling GPS-Denied Autonomy Across Embodiments”**. IEEE International Conference on Advanced Robotics (ICAR) 2025. [DOI](#).
4. M. Harms, **M. Kulkarni**, N. Khedekar, M. Jacquet, K. Alexis. **“Neural Control Barrier Functions for Safe Navigation”**. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2024. [DOI](#).
5. **M. Kulkarni** and K. Alexis. **“Reinforcement Learning for Collision-free Flight Exploiting Deep Collision Encoding”**. IEEE International Conference on Robotics and Automation (ICRA) 2024. [DOI](#).
6. M. Dharmadhikari, P. De Petris, **M. Kulkarni**, N. Khedekar, H. Nguyen, A.E. Stene, E. Sjøvold, K. Solheim, Bente Gussiaas, and Kostas Alexis. **“Autonomous Exploration and General**

- Visual Inspection of Ship Ballast Water Tanks using Aerial Robots.”**, IEEE International Conference on Advanced Robotics (ICAR) 2023. *Winner - Best Paper Award*. [DOI](#).
7. [M. Kulkarni](#) and K. Alexis, “**Task-driven Compression for Collision Encoding based on Depth Images**”. International Symposium on Visual Computing (ISVC) 2023. [DOI](#).
 8. [M. Kulkarni](#), H. Nguyen, and K. Alexis. “**Semantically-enhanced Deep Collision Prediction for Autonomous Navigation using Aerial Robots**”. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2023. [DOI](#).
 9. M. Dharmadhikari, P. De Petris, H. Nguyen, [M. Kulkarni](#), N. Khedekar and K. Alexis, “**Manhole Detection and Traversal for Exploration of Ballast Water Tanks using Micro Aerial Vehicles**”, International Conference on Unmanned Aircraft Systems (ICUAS) 2023, [DOI](#).
 10. N. Khedekar, [M. Kulkarni](#) and K. Alexis, “**MIMOSA: A Multi-Modal SLAM Framework for Resilient Autonomy against Sensor Degradation**”, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2022. [DOI](#).
 11. P. De Petris, H. Nguyen, M. Dharmadhikari, [M. Kulkarni](#), N. Khedekar, F. Mascarich, and K. Alexis. “**RMF-Owl: A Collision-Tolerant Flying Robot for Autonomous Subterranean Exploration.**”, International Conference on Unmanned Aircraft Systems (ICUAS) 2022. [DOI](#).
 12. [M. Kulkarni](#), M. Dharmadhikari, M. Tranzatto, S. Zimmermann, V. Reijgwart, P. De Petris, H. Nguyen, N. Khedekar, C. Papachristos, L. Ott, R. Siegwart, M. Hutter, and K. Alexis, “**Autonomous Teamed Exploration of Subterranean Environments using Legged and Aerial Robots**”, IEEE International Conference on Robotics and Automation (ICRA) 2022. [DOI](#). *Winner - Outstanding Deployed Systems Paper Award*.
 13. P. De Petris, H. Nguyen, [M. Kulkarni](#), F. Mascarich and K. Alexis, “**Resilient Collision-tolerant Navigation in Confined Environments**”, 2021 IEEE International Conference on Robotics and Automation (ICRA), 2021. [DOI](#).
 14. [M. Kulkarni](#), H. Nguyen, and K. Alexis, “**The Reconfigurable Aerial Robotic Chain: Shape and Motion Planning**”. IFAC World Congress, 2020. [DOI](#).

BOOK CHAPTERS AND WHITE PAPERS

1. [M. Kulkarni](#) et. al. “**Aerial Field Robotics**”, Encyclopedia of Robotics. [DOI](#).
2. M. Mittal et. al. “**Isaac Lab: A GPU Accelerated Simulation Framework For Multi-Modal Robot Learning**”. [DOI](#).
3. M. Dharmadhikari et. al. “**The Unified Autonomy Stack: Toward a Blueprint for Generalizable Robot Autonomy**”. [Link](#).

TALKS AND LECTURES

1. **PX4 Developer Summit 2025** - “From Pixels To Propellers: Sim2Real Control and Vision-based Navigation”. [Link](#)
2. **Invited Lecture: Worcester Polytechnic Institute**: “Reinforcement learning for control and navigation of aerial robots”
3. **Tutorial: Learning-oriented Simulation for Aerial Robots, SSCI 2025** - “Aerial Gym 2.0: Isaac Gym-based Massively Parallelized Simulation for Efficient Aerial Robot Learning”

ACADEMIC SERVICE

1. **Associate Editor**: IEEE/RSJ IROS
2. **Reviewer (Journals)**: RA-L, RAM, IJRR, T-RO, T-FR
3. **Reviewer (Conferences)**: ICRA, IROS, ICUAS, ICAR

SUPERVISION

Master’s Thesis

1. Theodor Johannes Line Forgaard - 2023

2. Ingvild Christoffersen Fleisje - 2023
3. Mathias Skogli - 2024
4. Ask Espensønn Øren - 2024

OPEN SOURCE CONTRIBUTIONS

1. **Unified Autonomy Stack** - a field-tested autonomy architecture commanding a diverse set of robots. [GitHub Website](#)
2. **Aerial Gym Simulator** - massively parallelized aerial robot simulator based on NVIDIA Isaac Gym. [GitHub Website](#).
3. **Semantically-enhanced Variational Autoencoder**. [GitHub](#).
4. **GSOC 2020: Sensor Data Visualization** - Open Robotics. [Link](#).
5. **Simulation Models** - Team CERBERUS - DARPA Subterranean Challenge Simulator. [GitHub](#).
6. **SuperMegaBot Simulator** - Team CERBERUS Roving Robot [GitHub](#).

PROGRAMS, INTERNSHIPS AND EXPERIENCE

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|---|-----------------------------|
| 1. Nordic Probabilistic AI School - ProbAI | <i>June 2023</i> |
| 2. Google Summer of Code 2020, Open Robotics | <i>May 2020 - Sep 2020</i> |
| 3. Visiting Scholar, University of Nevada, Reno | <i>June 2019 - Jan 2020</i> |
| 4. Summer Research Intern, CSIR-CEERI Pilani, India | <i>May 2018 - July 2018</i> |

MEDIA COVERAGE

1. **IEEE Spectrum**: Video Friday
 - (a) Reinforcement Learning for Collision-free Flight Exploiting Deep Collision Encoding. [Link](#)
 - (b) Autonomous Teamed Exploration of Subterranean Environments using Legged and Aerial Robots. [Link](#)
 - (c) Semantically-enhanced Deep Collision Prediction for Autonomous Navigation using Aerial Robots. [Link](#)
 - (d) DARPA SubT Finals: Robot Operator Wisdom. [Link](#)
2. **The Washington Post Magazine**: “The Pentagon’s \$82 Million Super Bowl of Robots”. [Link](#)
3. **Teknisk Ukeblad**: “Seier for NTNU-basert robotmiljø: Bedre enn både Nasa og MIT”. [Link](#)
4. **Gemini.no** “Vant 17 millioner med undergrunnsroboter”. [Link](#)
5. **GazeboSim Community**: GSOC 2020: Sensor Data Visualisation. [Link](#)
6. **BITS R&D**: Thesis at The University of Nevada, Reno. [Link](#)

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

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|--|----------------------------|
| 1. Chief Coordinator , Aerodynamics Club, BITS Pilani | <i>Mar 2018 - May 2019</i> |
| 2. Sub-Coordinator , Electronics and Robotics Club, BITS Pilani | <i>Apr 2018 - May 2019</i> |
| 3. Electronics Team Lead , Hyperloop India | <i>Mar 2018 - Jan 2019</i> |

TEACHING AND RESEARCH EXPERIENCE

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|--|----------------------------|
| 1. Teaching Assistant - Computer Programming (BITS Pilani) | <i>Jan 2020 - May 2020</i> |
| 2. Instructor - Intermediate Robotics (CTE, BITS Pilani) | <i>Jan 2019 - May 2019</i> |
| 3. Instructor - Introduction to Robotics (CTE, BITS Pilani) | <i>Aug 2018 - Dec 2018</i> |

SKILLS AND PROFICIENCIES

1. **AI/ML**: Deep Learning, Reinforcement Learning
2. **Robotics**: Path Planning, Multi-robot coordination, Field Robotics, Sim2Real Transfer
3. **Robotics Middleware and Tools**: ROS, ROS 2, PX4
4. **Simulation**: NVIDIA Isaac Gym/Sim/Lab, Gazebo, Blender
5. **Programming**: Python, C++, PyTorch, NVIDIA Warp, Docker, CI/CD
6. **Mechanical**: SOLIDWORKS, PTC Creo, Autodesk Fusion, Onshape
7. **Hardware**: Rapid prototyping, PCB Design (interfacing), Autopilot Systems
8. **Licenses**: Remote UAS Pilot - A1,A2,A3 (EASA)